17.04.25

OPENCAMPUS.sh

Introduction to Data Science and Machine Learning

INTRODUCTION

- Introduction
- Organizational information
- Course structure
- What is Data Science?
- Tools for programming

PERSONAL INTRODUCTION

OPENCAMPUS.sh

- Nonprofit organization which oversees a variety of initiatives
- Offering a wide range of educational opportunities, support, and networking for entrepreneurs, creatives, and anyone curious, regardless of age, educational background, or origin
- The services are open to everyone and mostly free.
- The goal is to support the entrepreneurial landscape, promote creative change processes, and contribute to innovative and sustainable future development.









starter kitchen.de



COBL

COZY WORKING, CULTURE & EVENTS





KOSMOS
by opencampus.sh











FABLAB KIEL

SEEd | Social Entrepreneurship Education

#seed17







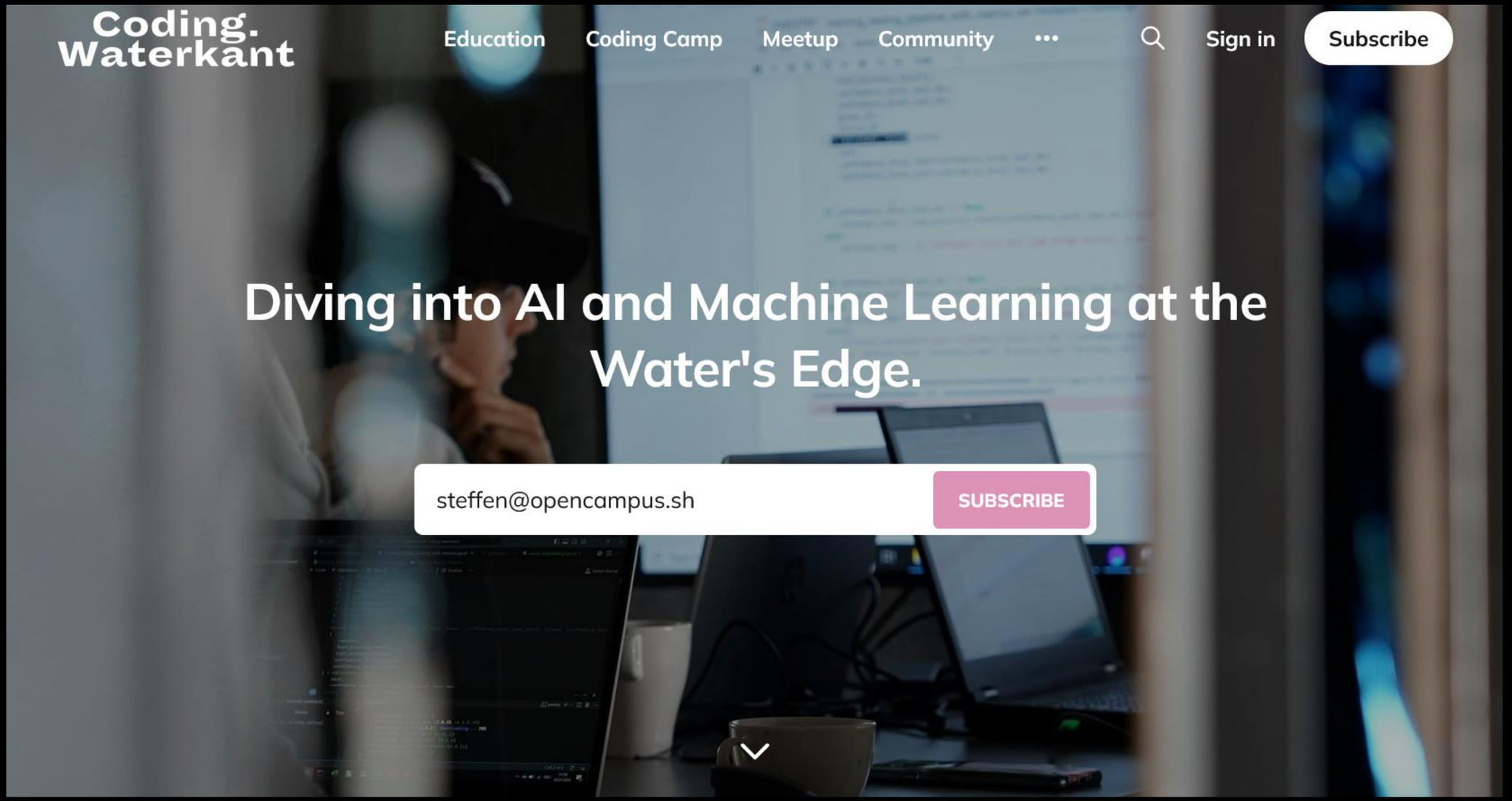


Design Thinking

Business/
Project Modelling

Pitching/
Präsentationen


Visual Science
Communication
Research
WAS SIND OZEANE WERT?
HOW MUCH ARE OCEANS WORTH?
WATERKANT
FESTIVAL
June 19-20



https://coding.waterkant.sh



Coding. Waterkant 2025

Coding.Waterkant 2025

July 07 - 11

- Work for four consecutive days:
  - on your own machine learning project or
  - support others in their project
- Take part in workshops
- Get input and feedback by invited experts
- Present your work to a larger audience.
- Take educational leave
- Get accommodation on site

Meetup

Events suchen | Kiel, DE

Neue Gruppe starten - 30% Rabatt!

PRO Kostenlos ausprobieren

Vernetzungen

Im Rahmen der ständigen Verbesserungen unserer Plattform haben wir die E-Mail-Adresse deiner Gruppe in coding-waterkant-announce@messages.meetup.com geändert. Du kannst dies in den optionalen Einstellungen deiner Gruppe ansehen.



Coding.Waterkant

★★★★½ (126)

Neu: Event-Feedbackübersicht

Tippe auf die Sterne, um alle deine Eventbewertungen in der Übersicht anzuzeigen.

Weitere Informationen

Kiel, Deutschland

981 Mitglieder · Öffentliche Gruppe

Organisiert von **opencampus.sh** and **6 others**

Teilen:

Info | Events | Mitglieder | Fotos | Diskussionen | Mehr

Event erstellen | Gruppe verwalten

Über uns

Our meetup is organized with support of opencampus.sh and the Digitale Wirtschaft Schleswig-Holstein (DiWiSH).

Mehr lesen



Organizers

opencampus.sh and **6 others**

Nachricht

Members (981)

Alles ansehen



ORGANIZATIONAL MATTERS

Attendance Registration:

Online

- **Use your full names in the zoom meetings!**
- **Turn your camera on!**

Presence

- **Scan the QR-Code if you participate in presence**

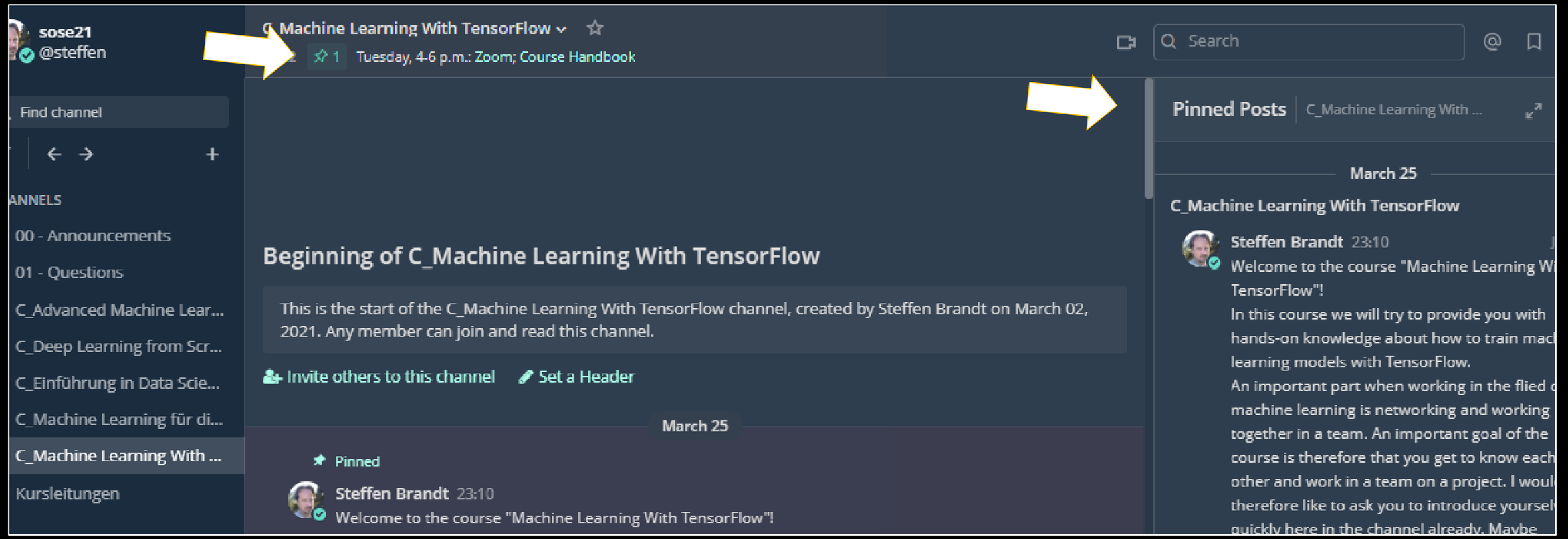
Please write us if you will not go on with the course!

MATTERMOST CHAT

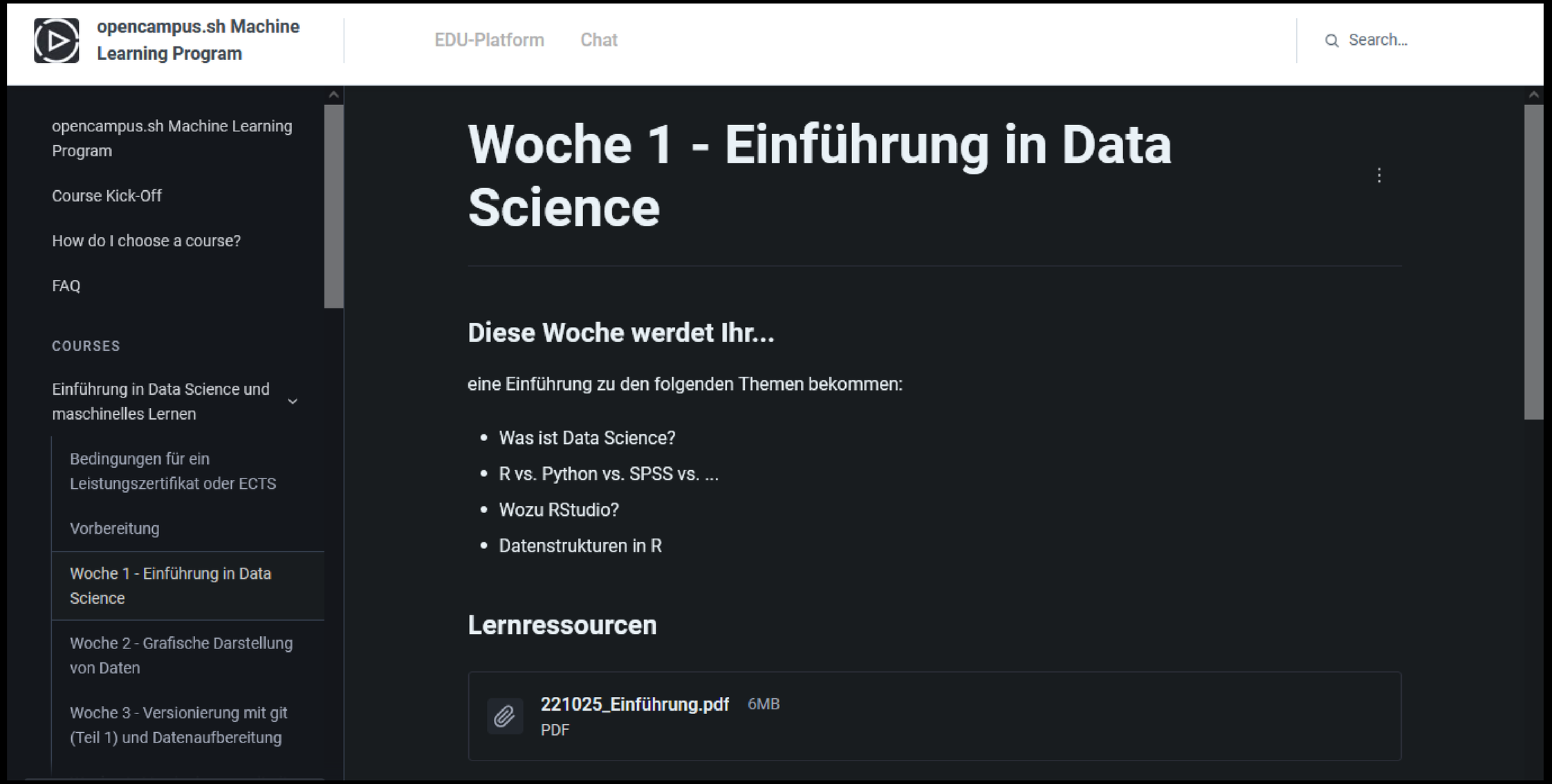
- Complete your profile in the Mattermost chat with your full name and a photo.

- Please, always ask any questions to us in the course chat!!

MATTERMOST CHAT



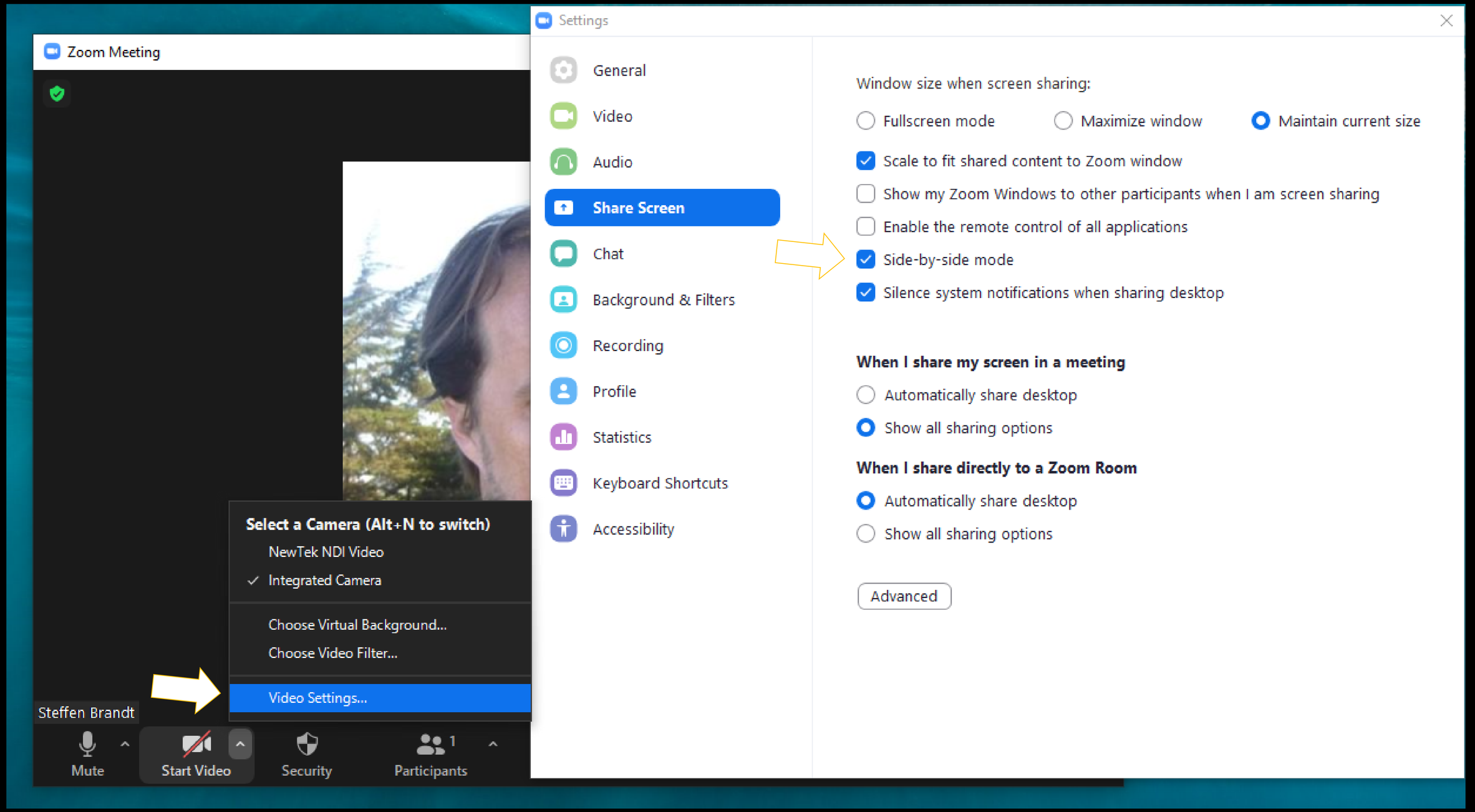
COURSE HANDBOOK



ZOOM

- Try the different viewing modes:
  - Gallery View/ Active Speaker
  - Split Screen/ Full Screen Mode

- Maybe watch this video to get an idea: https://www.youtube.com/watch?v=v3IPAbpVjd4


Zoom Meeting
Select a Camera (Alt+N to switch)
NewTek NDI Video
Integrated Camera
Choose Virtual Background...
Choose Video Filter...
Video Settings...
Steffen Brandt
Mute
Start Video
Security
Participants
1
Settings
General
Video
Audio
Share Screen
Chat
Background & Filters
Recording
Profile
Statistics
Keyboard Shortcuts
Accessibility
Window size when screen sharing:
Fullscreen mode
Maximize window
Maintain current size
Scale to fit shared content to Zoom window
Show my Zoom Windows to other participants when I am screen sharing
Enable the remote control of all applications
Side-by-side mode
Silence system notifications when sharing desktop
When I share my screen in a meeting
Automatically share desktop
Show all sharing options
When I share directly to a Zoom Room
Automatically share desktop
Show all sharing options
Advanced

BREAKOUTS

For Online Participants:

Take a screenshot of the task before joining breakouts!

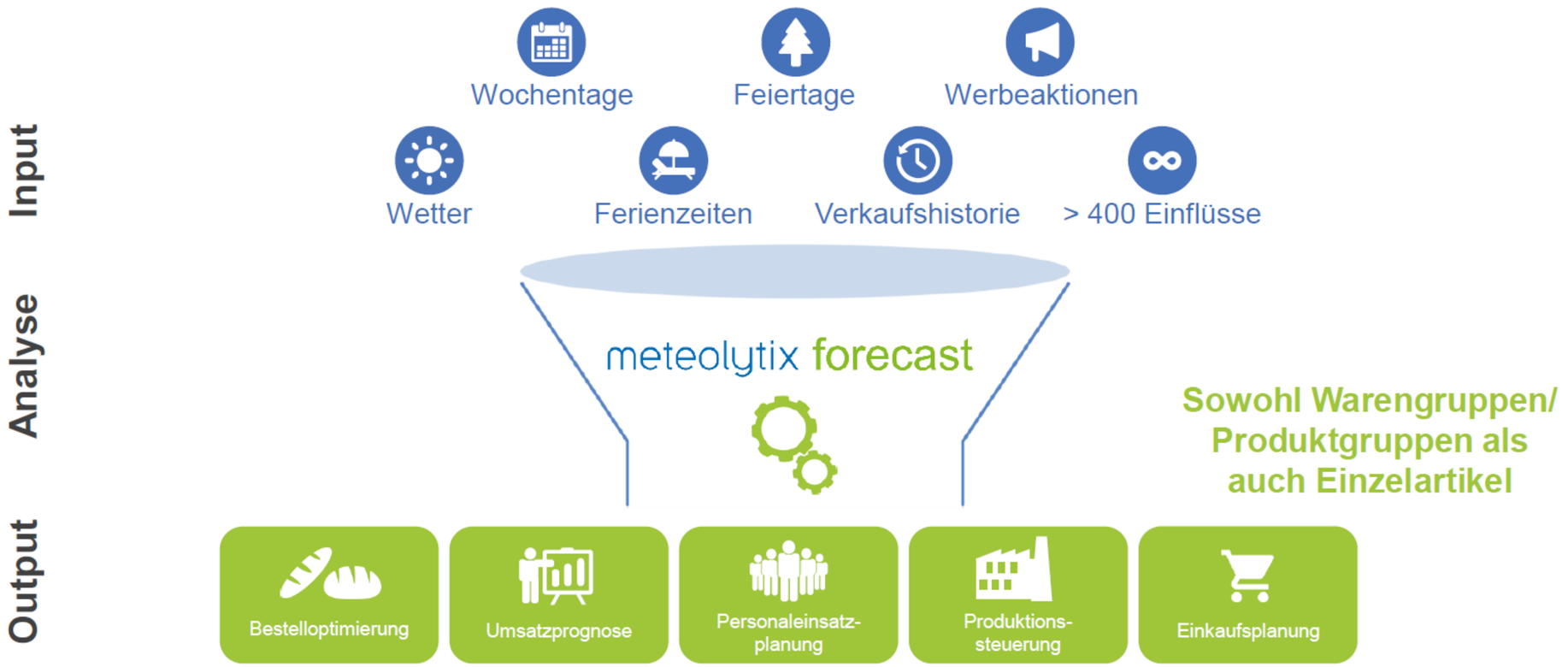
SALES FORECAST PROJECT

- Real sales data from various product groups of a bakery branch over a period of approximately 3 years

- Weather data for the corresponding period (and beyond)

meteolytix forecast analysiert die Datenzusammenhänge von mehr als 400 Einflussfaktoren und liefert Absatzprognosen für viele Einsatzfelder.

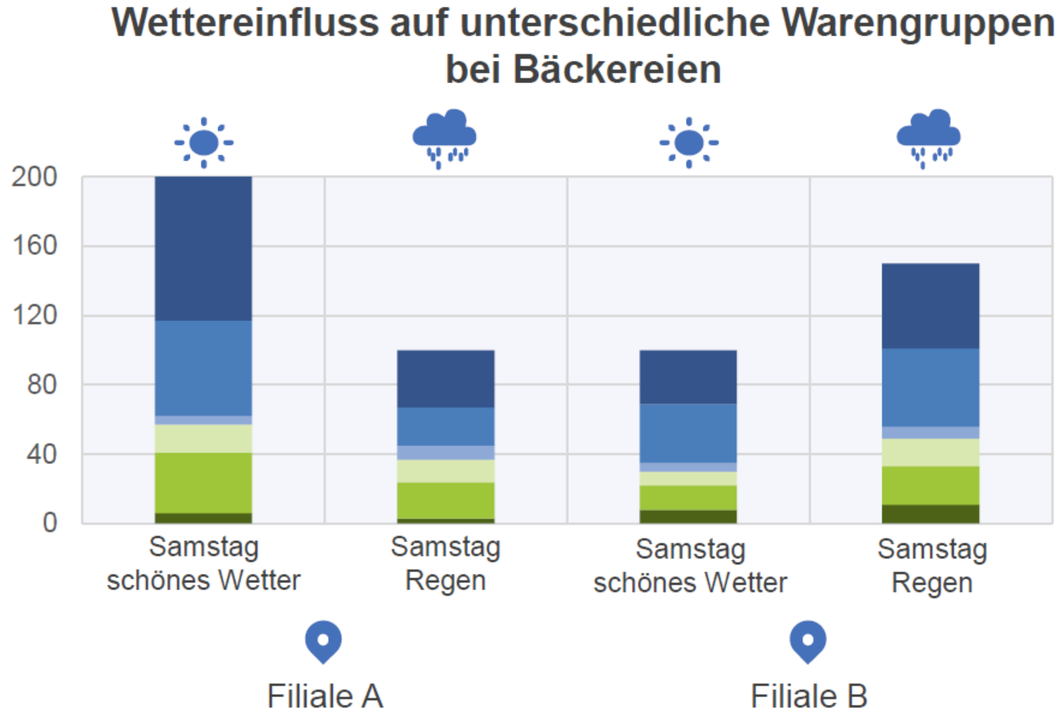
WAS WIR MACHEN

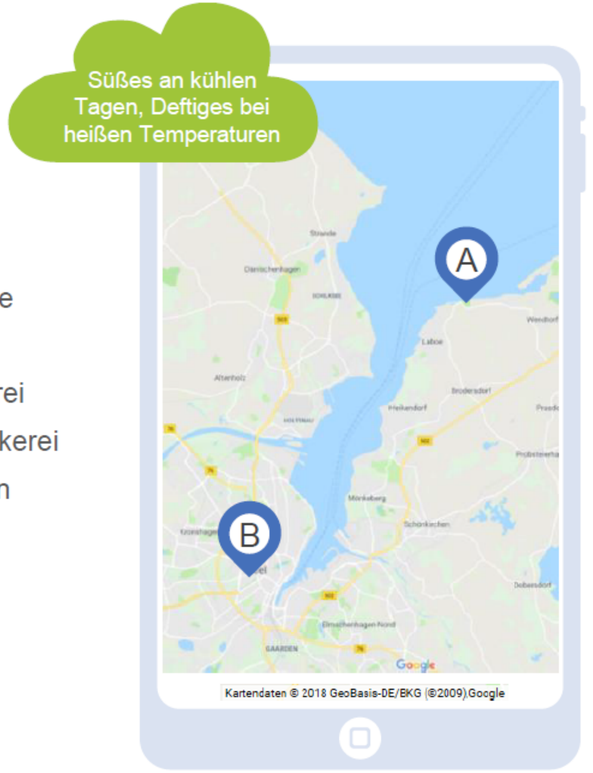


© 2019 *meteolytix GmbH*

Die Stärke des Wettereffekts variiert von Ort zu Ort und wird jeweils filialindividuell berücksichtigt.

WAS WIR MACHEN

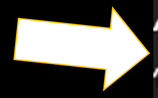




PROJECT GOALS

- Programming with AI support in VS Code
- Working in a team with version control (Git)
- Preparation of variables
- Use of graphical representations
- Use of regressions to identify relevant variables
- Handling missing values
- Optimization of a neural network

Definition of the Teams



17.04.2025	Introduction to Data Science	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
24.04.2025	Data Visualization	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
08.05.2025	Version Control with Git and GitHub	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
15.05.2025	Data Preparation	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
22.05.2025	Overfitting and Model Evaluation	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
05.06.2025	Introduction to Machine Learning	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
12.06.2025	Neural Networks	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
19.06.2025	Handling Missing Values	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
26.06.2025	Time Series Analysis	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	
03.07.2025	Project Presentations	
16:00 - 18:00	Starterkitchen, Kuhnkestr. 6, 24118 Kiel + ONLINE	

„Education is the only thing that humans try to get less out of than what they pay for.“

FIRST BREAKOUT

- 15 Minutes
- Present yourself
- Discussion Question:
  How is Artificial Intelligence defined?
  - Try to agree on a definition
  - Share your definition in the course channel in Mattermost

WHAT IS AI?

- Chess computer
- Google Maps
- ChatGPT

“What's the difference between data science, machine learning, and artificial intelligence?”

http://varianceexplained.org/r/ds-ml-ai

Data science produces insights.

Machine learning produces predictions.

Artificial intelligence produces actions.

DATA SCIENCE

Application of statistical analysis methods to generate results for a report or article.

- **Descriptive statistics**
- **Statistical inferences**
- **Data visualization**
- **Design of experiments**

MACHINE LEARNING

The use of statistical optimization techniques to estimate model parameters for a prediction function validated with unseen data.

Typically, model performance is continuously improved by incorporating new data, allowing the system to adapt and refine its predictions over time.

- **Predicting future states**
- **Inferring unobserved states**
- **Estimating current states**

ARTIFICIAL INTELLIGENCE

„an autonomous agent executes or recommends actions“

(Poole, Mackworth, & Goebel, 1998)

„Artificial intelligence (AI) refers to systems that display intelligent behaviour by analysing their environment and taking actions – with some degree of autonomy – to achieve specific goals. “

(European Commission, 2018)

„ By Artificial Intelligence, we understand highly developed software systems that are capable of learning and can be trained to handle complex tasks.“

(AI Strategy of Schleswig-Holstein, 2019)

TOOLS TO CREATE AND EXECUTE CODE

- Programming language + function libraries
- Execution environment
- Development environment

PROGRAMMING LANGUAGE: PYTHON

- Open source, huge community
  → large amounts of existing example code
  → very good AI support
- Industry and (slowly getting) academia standard for data analysis
- Primary ML language
- Fastest-growing language with wide application range including productive settings
- Extensive function libraries for:
  - Web Applications (z.B. Django und Flask)
  - Data Analysis (NumPy, Pandas)
  - Visualizations (Matplotlib, Seaborn)
  - Neural Networks (TensorFlow, PyTorch)

PYTHON CHARACTERISTICS

- Interpreter language (executes line-by-line, similar to a calculator)
- Console-based (no GUI)
- Evolved from software engineering (vs. R from statistics)

EXECUTION ENVIRONMENT:

Jupyter Notebook:

- **Browser-based execution of code via a GUI**
- **Popular in ML and Data Science**

Python Script:

- **Execution of code from a text file**
- **Used particularly in production**

IP[y]: Notebook spectrogram Last Checkpoint: a few seconds ago (autosaved)

IPython (Python 3)

File Edit View Insert Cell Kernel Help

Code Cell Toolbar: None

Simple spectral analysis

An illustration of the Discrete Fourier Transform using windowing, to reveal the frequency content of a sound signal.

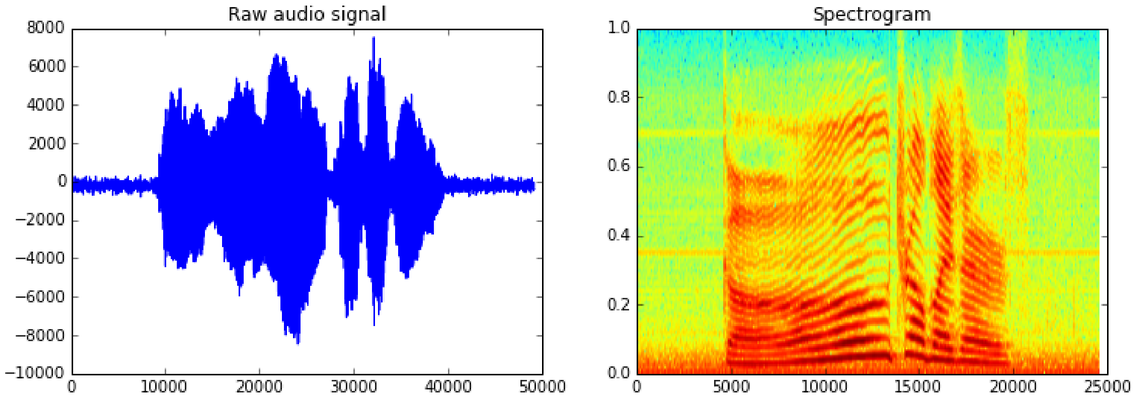
$$X_k = \sum_{n=0}^{N-1} x_n e^{-\frac{2\pi i}{N} kn} \qquad k = 0, \ldots, N-1$$

We begin by loading a datafile using SciPy's audio file support:

In [1]:
```
from scipy.io import wavfile
rate, x = wavfile.read('test_mono.wav')
```

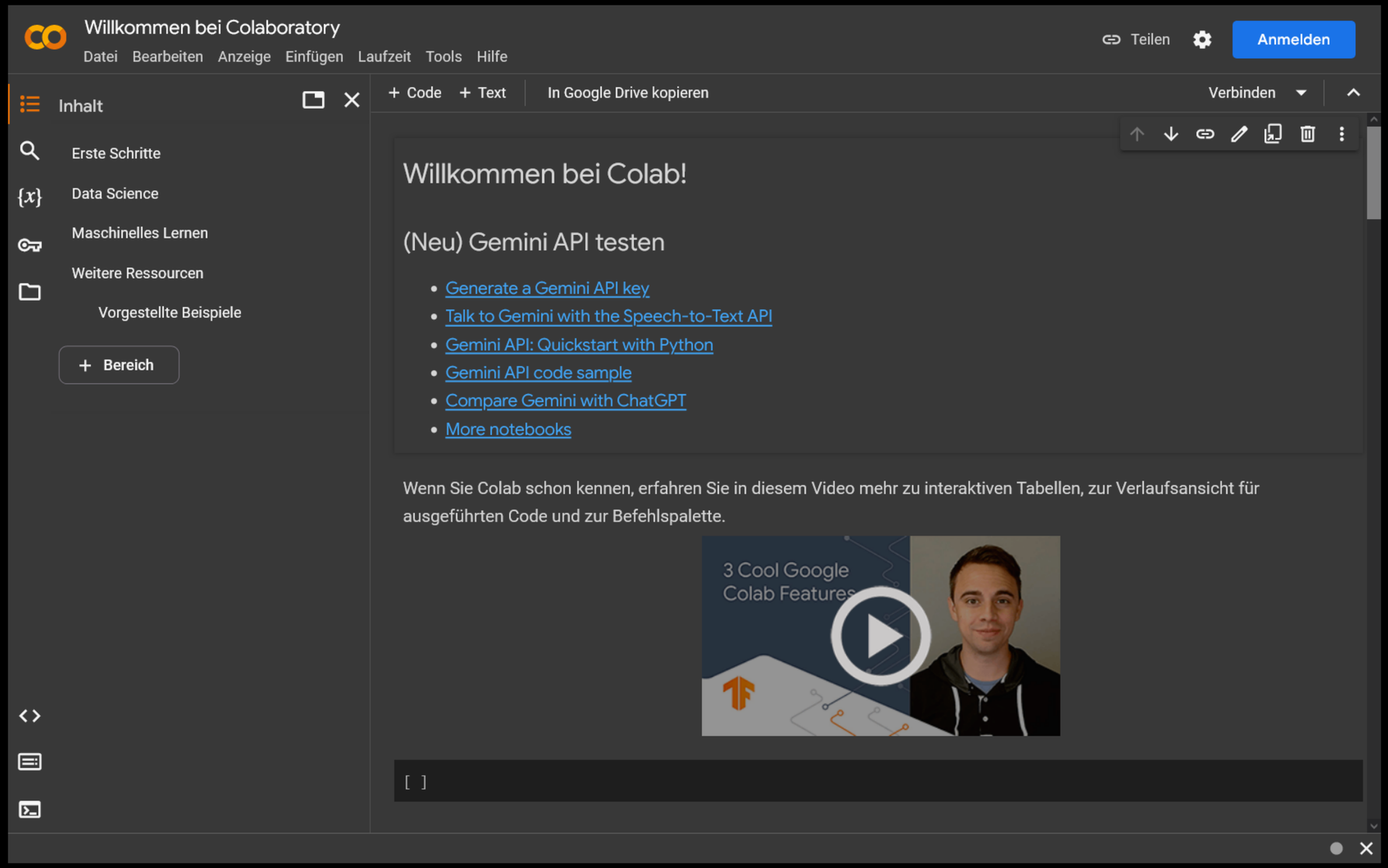
And we can easily view its spectral structure using matplotlib's builtin `specgram` routine:

In [2]:
```
%matplotlib inline
from matplotlib import pyplot as plt
fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 4))
ax1.plot(x); ax1.set_title('Raw audio signal')
ax2.specgram(x); ax2.set_title('Spectrogram');
```



COLAB

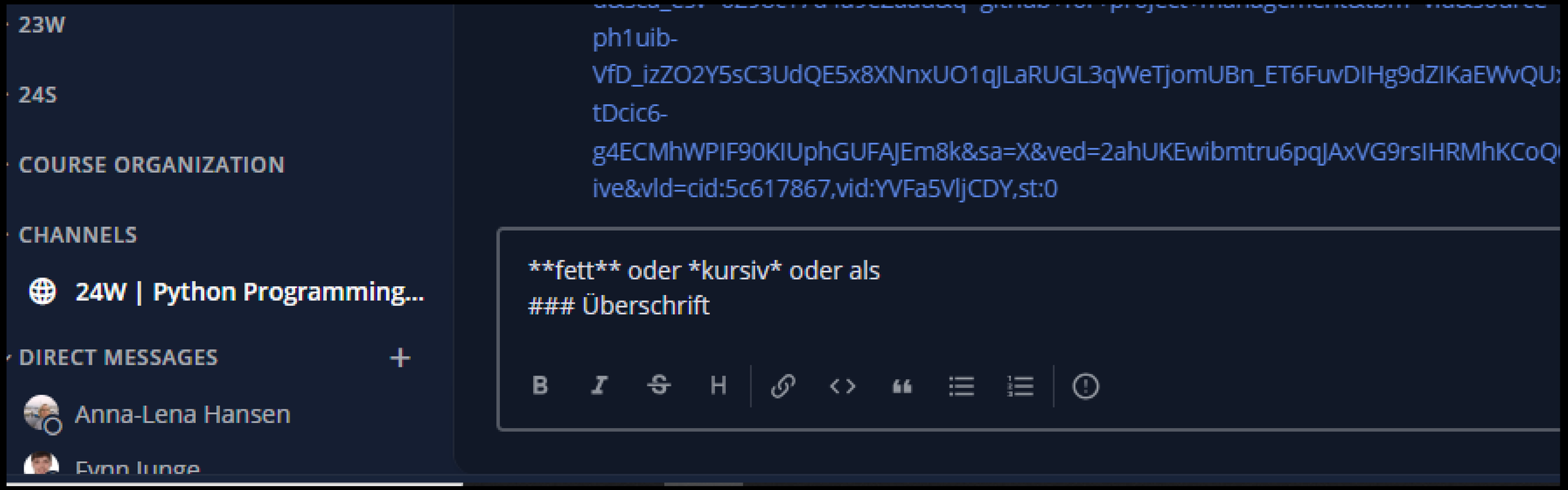
https://colab.research.google.com/



CHARACTERISTICS OF JUPYTER NOTEBOOKS

- Open Source application, run in a web browser or locally
- Convenient editing and execution of code in the form of code cells
- Display of results directly in the notebook below the executed code cell – including graphical output
- Insertion of formatted comments and explanations via Markdown cells

MARKDOWN IN MATTERMOST



EXAMPLES FOR MARKDOWN SYNTAX

Markdown	Preview
`**bold text**`	**bold text**
`*italicized text*` or `_italicized text_`	*italicized text*
`` `Monospace` ``	`Monospace`
`~~strikethrough~~`	~~strikethrough~~
`[A link](https://www.google.com)`	A link
`![An image](https://www.google.com/images/rss.png)`	

DEVELOPMENT ENVIRONMENT: VISUAL STUDIO CODE (VS CODE)

- Open-source development environment that supports the creation of programming code
- Standard development environment for numerous programming languages in software development
- Many extensions and additional features available via plugins

TOOLS TO CREATE AND EXECUTE CODE

Software:

- **Programming Language**
  → Python und Python Packages
- **Execution Environment**
  → Python Environments or Jupyter Notebooks
- **Development Environment**
  → VS Code

TOOLS TO CREATE AND EXECUTE CODE

Hardware:

- **Previously, typically only the local computer**
- **Due to increasing computational requirements, code is now often executed in the cloud**
- **Development environments are also increasingly used in the cloud**

OUR APPROACH

- Local installation of the VS Code development environment
- Execution of the code in an environment in the cloud using GitHub Codespaces
- Linking the local development environment with the cloud-based execution environment

WHY GITHUB CODESPACES?

- **No local Python installation required**
  → Avoids potential compatibility and configuration issues across different operating systems
- **Consistent Execution Environments**
  → Everyone uses the same Python versions and code examples work the same for all
- **Integration with GitHub and GitHub Copilot**
  → No additional setup required
- **Live Sharing**
  → Share development environments for real-time collaboration
- **Automatc Backups**

EXAMPLE USAGE

1. Create a GitHub repository (a cloud-based project workspace)
2. Add a GitHub Codespace to this repository
3. Open the Codespace with your local VS Code

steffen74

Type / to search

Overview | Repositories 16 | Projects | Packages | Stars 18



Steffen Brandt

steffen74

Edit profile

23 followers · **1** following

Find a repository... | Type | Language | Sort | New

workshop-ressources

Public

Example data sets and programm code for the workshop at the EALTA 2019

Jupyter Notebook Updated on Jun 28

Star

GenAI-powered-Literature-Analysis

Public

Forked from ApaydinK/GenAI-powered-Literature-Analysis

Python Updated on Jun 14

Star

bakery_sales_prediction

Public

Updated on May 9

Star

ConstitutionalAiTuning

Public

https://github.com/new

Create a new repository

A repository contains all project files, including the revision history. Already have a project repository elsewhere?
Import a repository.

Required fields are marked with an asterisk (*).

Repository template

No template

Start your repository with a template repository's contents.

Owner * **Repository name ***

steffen74 / test-repo

test-repo is available.

Great repository names are short and memorable. Need inspiration? How about shiny-carnival ?

Description (optional)

Public
Anyone on the internet can see this repository. You choose who can commit.

Private
You choose who can see and commit to this repository.

Initialize this repository with:

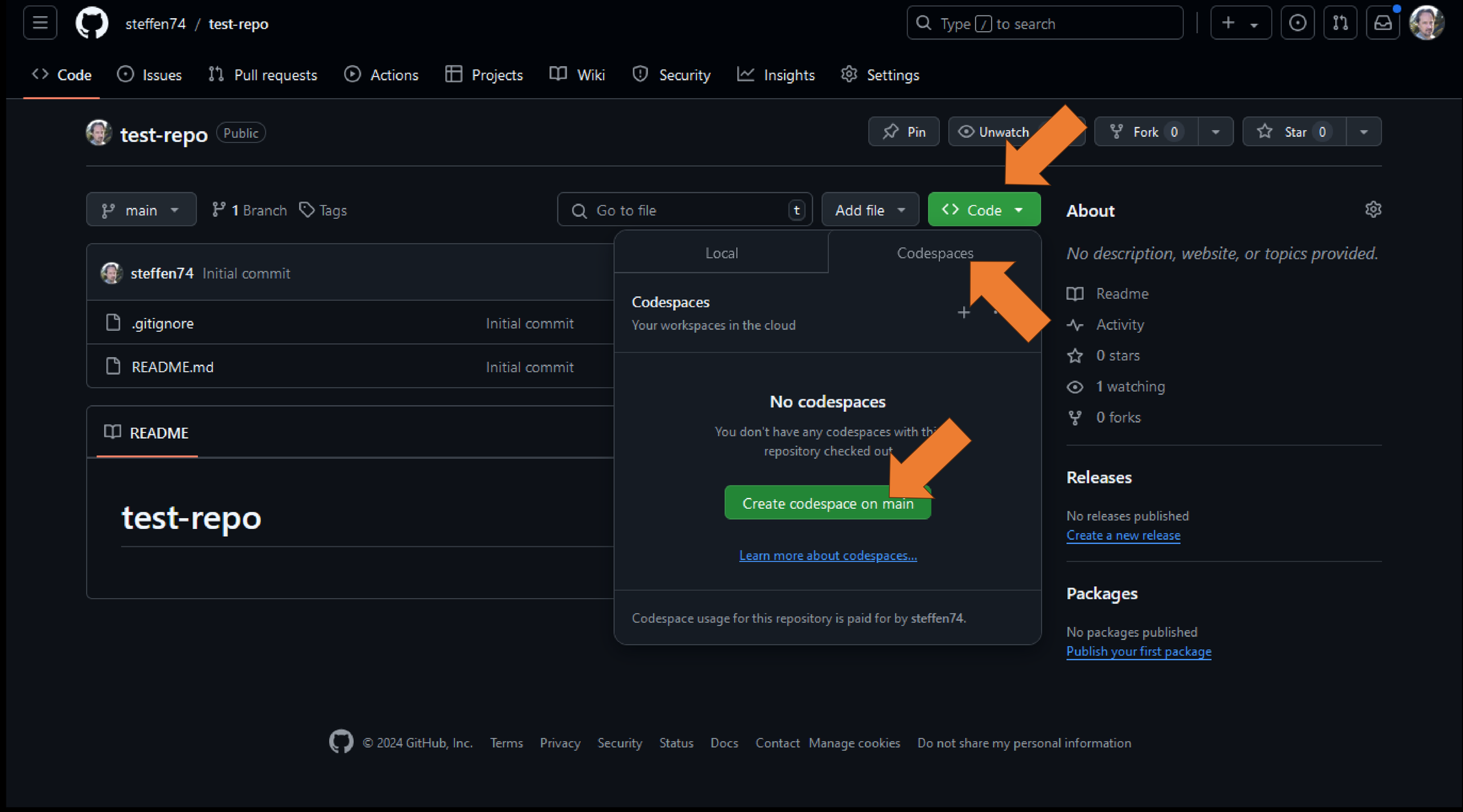
Add a README file
This is where you can write a long ... for your project. Learn more about READMEs.

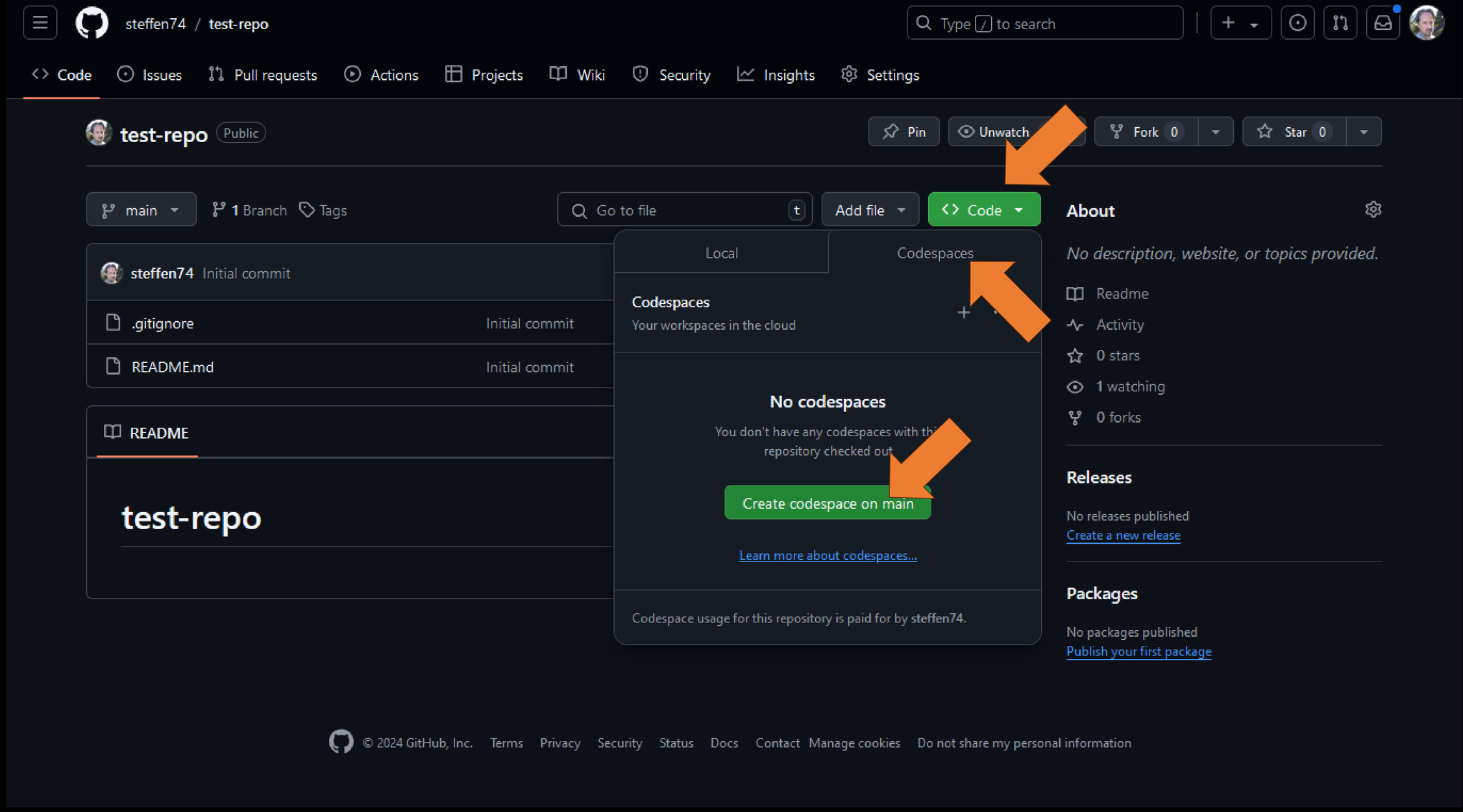
Add .gitignore

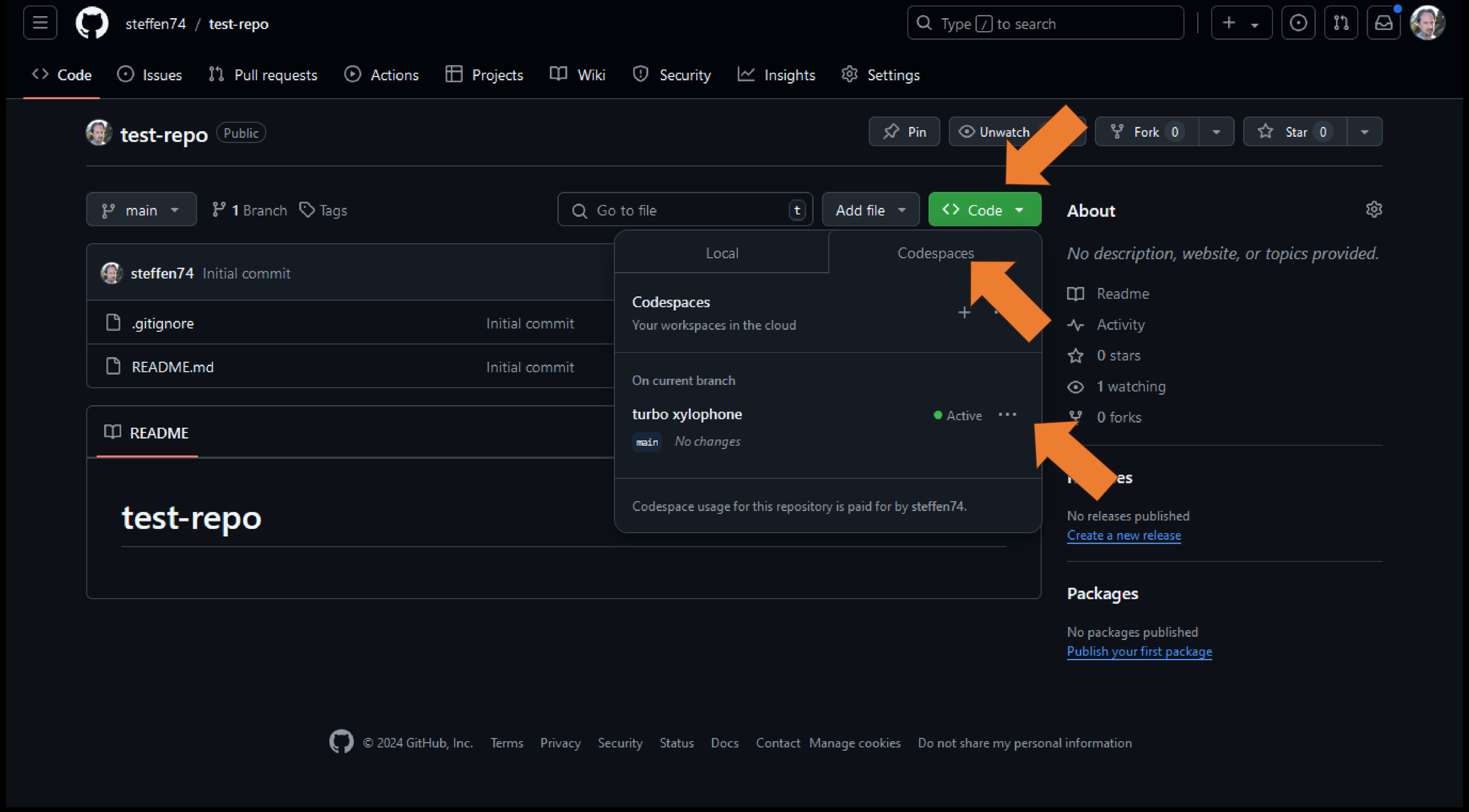
.gitignore template: Python

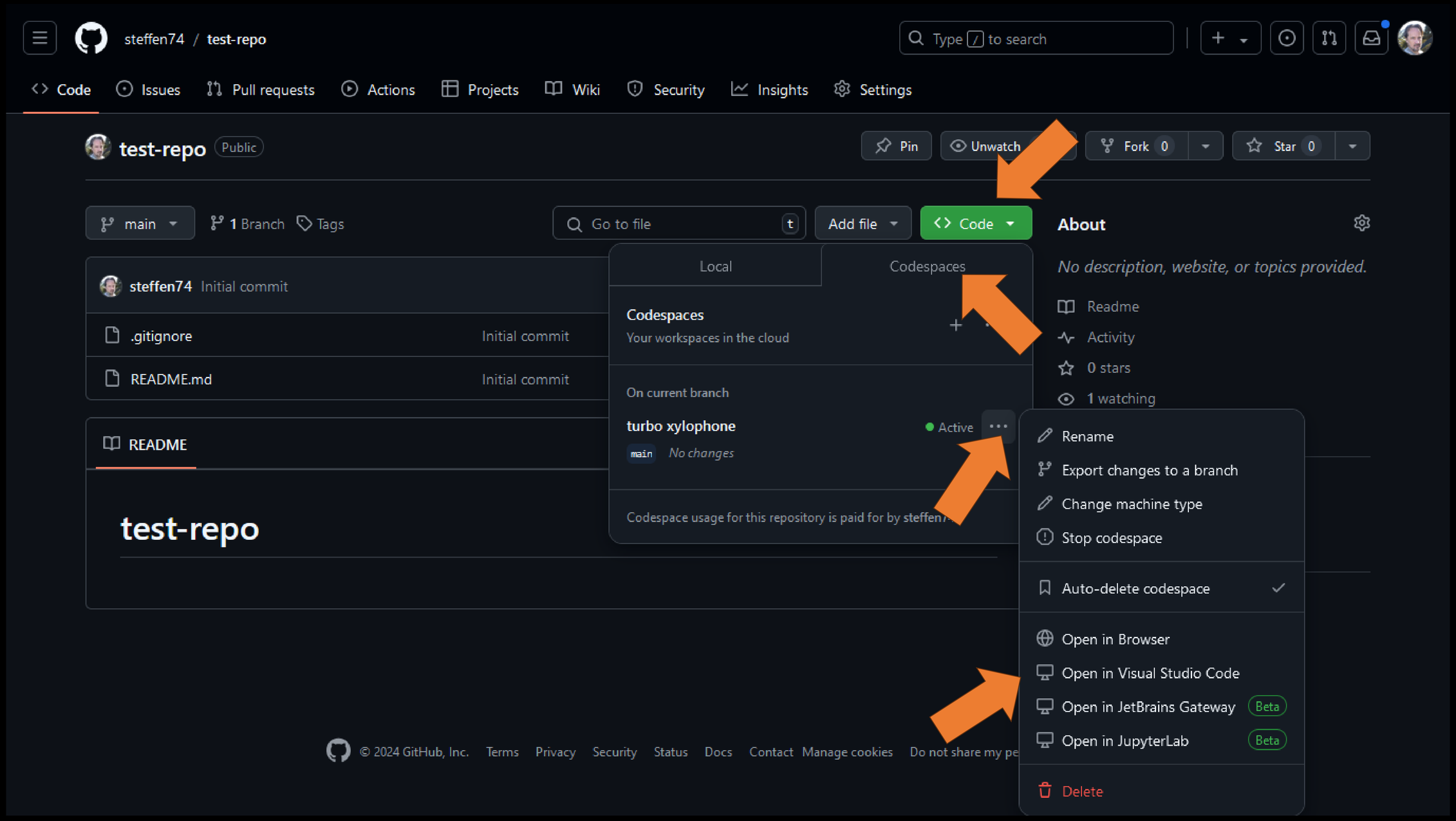
Choose which files not to track from a list of templates. Learn more about ignoring files.

Choose a license


steffen74 / test-repo
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
test-repo
Public
Pin
Unwatch
Fork 0
Star 0
main
1 Branch
Tags
Go to file
t
Add file
Code
steffen74 Initial commit
.gitignore Initial commit
README.md Initial commit
README
test-repo
Local
Codespaces
Codespaces
Your workspaces in the cloud
No codespaces
You don't have any codespaces with th
repository checked out
Create codespace on main
Learn more about codespaces...
Codespace usage for this repository is paid for by steffen74.
About
No description, website, or topics provided.
Readme
Activity
0 stars
1 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
© 2024 GitHub, Inc.
Terms
Privacy
Security
Status
Docs
Contact
Manage cookies
Do not share my personal information


steffen74 / test-repo
Type / to search
Code
Issues
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test-repo
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No description, website, or topics provided.
Readme
Activity
0 stars
1 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
© 2024 GitHub, Inc.
Terms
Privacy
Security
Status
Docs
Contact
Manage cookies
Do not share my personal information


steffen74 / test-repo
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
Settings
test-repo
Public
Pin
Unwatch
Fork 0
Star 0
main
1 Branch
Tags
Go to file
t
Add file
Code
Local
Codespaces
Codespaces
Your workspaces in the cloud
On current branch
turbo xylophone
Active
main
No changes
Codespace usage for this repository is paid for by steffen74.
steffen74 Initial commit
.gitignore
Initial commit
README.md
Initial commit
README
test-repo
About
No description, website, or topics provided.
Readme
Activity
0 stars
1 watching
0 forks
No releases published
Create a new release
Packages
No packages published
Publish your first package
© 2024 GitHub, Inc.
Terms
Privacy
Security
Status
Docs
Contact
Manage cookies
Do not share my personal information


steffen74 / test-repo
Type / to search
Code
Issues
Pull requests
Actions
Projects
Wiki
Security
Insights
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test-repo
Public
Pin
Unwatch
Fork 0
Star 0
main
1 Branch
Tags
Go to file
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Code
Local
Codespaces
Codespaces
Your workspaces in the cloud
On current branch
turbo xylophone
Active
main
No changes
Codespace usage for this repository is paid for by steffen
steffen74 Initial commit
.gitignore
Initial commit
README.md
Initial commit
README
test-repo
About
No description, website, or topics provided.
Readme
Activity
0 stars
1 watching
Rename
Export changes to a branch
Change machine type
Stop codespace
Auto-delete codespace
Open in Browser
Open in Visual Studio Code
Open in JetBrains Gateway
Beta
Open in JupyterLab
Beta
Delete
© 2024 GitHub, Inc.
Terms
Privacy
Security
Status
Docs
Contact
Manage cookies
Do not share my pe

BREAKOUT

Calculate the mean of the following seven numbers using Python:

```
1
2
3
4
5
6
9,87654321
```

SUPPORT TOOLS

- Code Completion
  (GitHub Copilot, Supermaven / Cursor Tab, ...)
- Specific Code Assistants
  (GitHub Copilot, Cursor Composer, Qodo Gen, ...)
- Chatbots
  (Claude, Gemini, ChatGPT, ...)
- Google
- Stackoverflow
- Dokumentation der Funktionen

LEARNING RESOURCES

- Watch this video on working with strings (16 minutes)
- Watch this snippet on working with numbers ("Numbers" and "Working With Numbers"; 5 minutes)
- Watch the first four chapters of this video on functions (12 minutes)
- Watch this video on working with lists in Python (18 minutes)
- Watch this video on setting up VS Code, specifically for Python and Data Science

TASKS

- Import the dataset "umsatzdaten_gekuerzt.csv", which you can download via the following link:
  https://raw.githubusercontent.com/opencampus-sh/einfuehrung-in-data-science-und-ml/main/wetter.csv
- Calculate the overall average temperature.
- Calculate the average temperature for the month of July.
- Compare whether the months of July and May differ significantly in their average temperature.

REGISTRATION FOR GITHUB COPILOT

- Register for GitHub Copilot as described here.

  As a student or teacher you get free access to Gitub Copilot Pro following the instructions given here.

Next week, we will go into more detail on possible ways of using Ai assistance.